# University of Baghdad
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**Title: Acute Respiratory Distress Syndrome.**

**Grade: 4th grade**

**Module: Respiratory Medicine Module**

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**Date:**

# Acute Respiratory Distress Syndrome

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# Acute Respiratory Distress Syndrome

- It is also called "Shock Lung“
- **Definition:** It is a syndrome in which there is pulmonary oedema without increment in the pulmonary capillary venous pressure or non cardiac pulmonary oedema

- Unlike the Lt.sided heart failure , in which there is pulmonary oedema due to increment in the Lt. atrial pressure which causes increment in the pulmonary venous pressure → oozing of fluids to the lung tissue → cardiac pulmonary oedema.

- While in ARDS, the pulmonary oedema is due to oozing of fluids through the capillaries without increment in pulmonary venous pressure which means that the defect in the respiratory membrane.

# *Causes*

There are many processes that precipitate ARDS.

The following are well known causes

1) **Pulmonary infection**:

A- Viral, e.g. H1N1 influenza, SARS,…

B- Mycoplasmal, ..

C- Bacterial,

D- Tuberculosis

**2) Injuries :**

A- Lung injury:

Either directly by bullet or blast …etc. or indirectly by fractures or hemolytic anemia.. etc.

B- Fractures

C- Head injury

D- Surgery – if prolonged- cause ARDS

**3) Septicemia & septic shock**: in fact all shock states can cause ARDS, that is why ARDS is called "Shock lung"

**4) Acute pancreatitis**

**5) Burn**

**6) Embolization :** either fatty embolus, pulmonary thrombo-embolus, air embolus, & amniotic fluid embolus..

** usually, embolization is due to small ambolus that blocks small tributary of pulmonary artery; but the patient may develop ARDS (unknown cause) or by big embolus → massive pulmonary embolization → ARDS

**7) Direct damage to the lung by acids** (due to vomiting or regurgitation) → ARDS

- **8) Drowning :** water is detrimental to the lung & can initiates ARDS

- **9) Hanging :** either suicidal or execution , if the patient –for example- survive after hanging himself → develop ARDS

- **10) Inhalation of chemicals & toxic materials :** as in the wars, which cause ARDS & serious respiratory problems to the soldiers, or it can happen to the workers (who works in water refinery system)where chlorine is used for water cleaning, but chlorine is very irritant to the lung & cause damage to the lung tissue →ARDS also happen in housewives who use detergent in large conc.

- **11) In cardio pulmonary bypass in cardiac surgery**

- **12) Multiple blood transfusion**

## 13. Toxic gases :

That cause this syndrome One of which is $O_2$ . Though $O_2$ is used in the treatment of this syndrome, it is an irritant gas if given in high conc. So it is mandated not to give high conc. for more than 24 hours, & this could happen in mechanical ventilators, if you breath into close circuit & getting pure $O_2$ (100% conc.), so if you need $O_2$, you should not use pure $O_2$ over 24hs because it may lead to this syndrome.

# Pathophysiology

Regardless of the cause of ARDS, it will lead to destruction of the lung tissue & this tissue contains endothelium of the blood vessels & the epithelium in the alveoli which constitute the respiratory membrane.

Hence, the process will lead to damage to both epithelium & the endothelium or to any one of them, & once the capillaries get damaged it will lead to oozing of oedema fluid to the alveolar spaces, this is actually a pulmonary oedema, (such thing can occur in left sided heart failure) & there will be oozing of fluid from the capillaries to the air-spaces of the lung tissue.

Hence, damage to the epithelial & endothelial layers due to inflammatory process of ARDS will lead to stiffness of the lung.

**Acute Phase**:

- Pulmonary oedema with normal vascular pedicle

Associated with no cardiomegaly or upper lobe blood diversion

When pulmonary vessels can be distinguished they are often constricted

- Septal lines usually absent because capillary leak occurs directly into alveolar spaces (cardiogenic pulmonary oedema)
- Progressive lung destruction and transition from alveolar to interstitial opacities

**Chronic phase**

- fibrosis
- focal emphysema

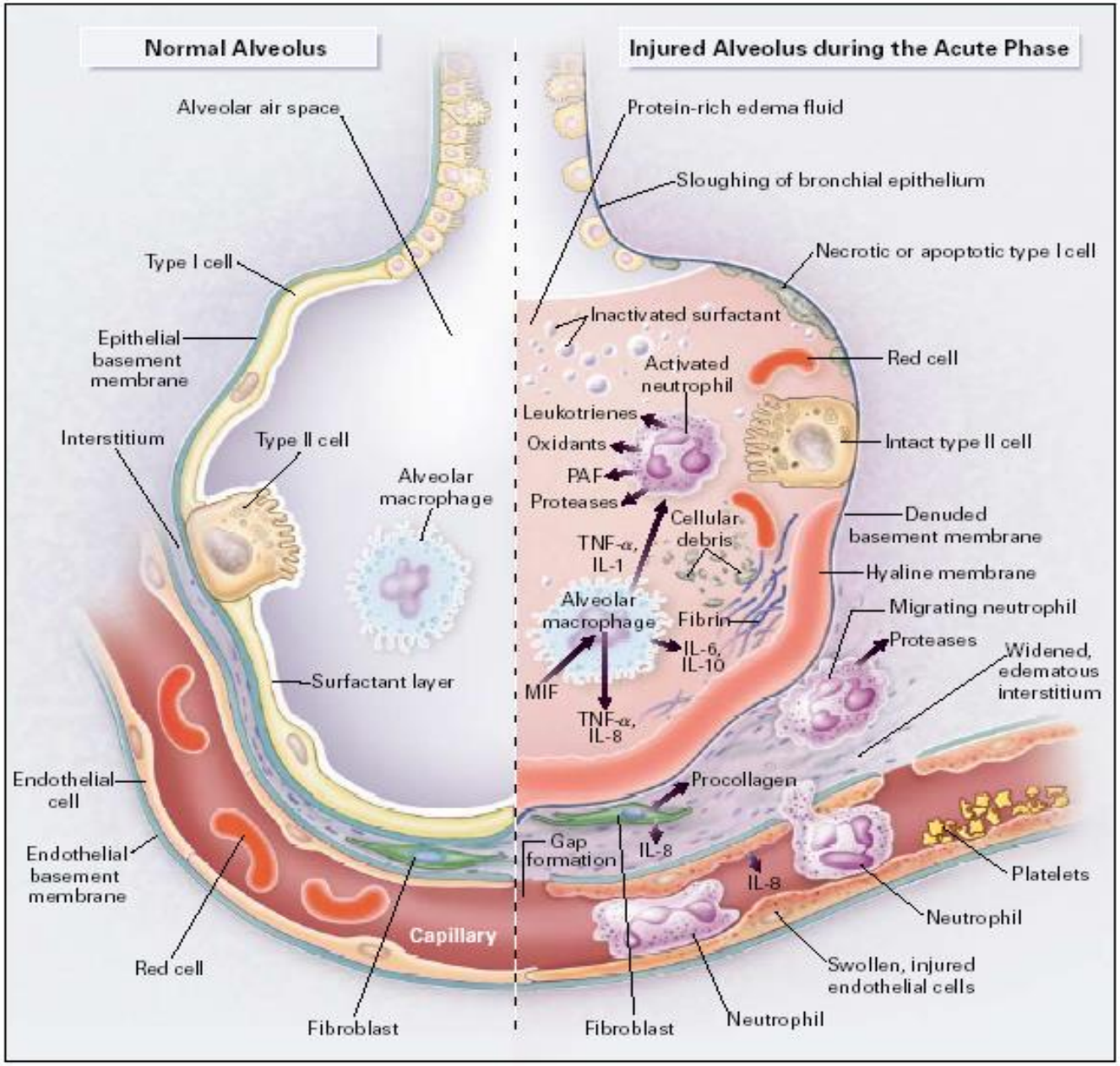
# Inflammatory Alveolar Injury

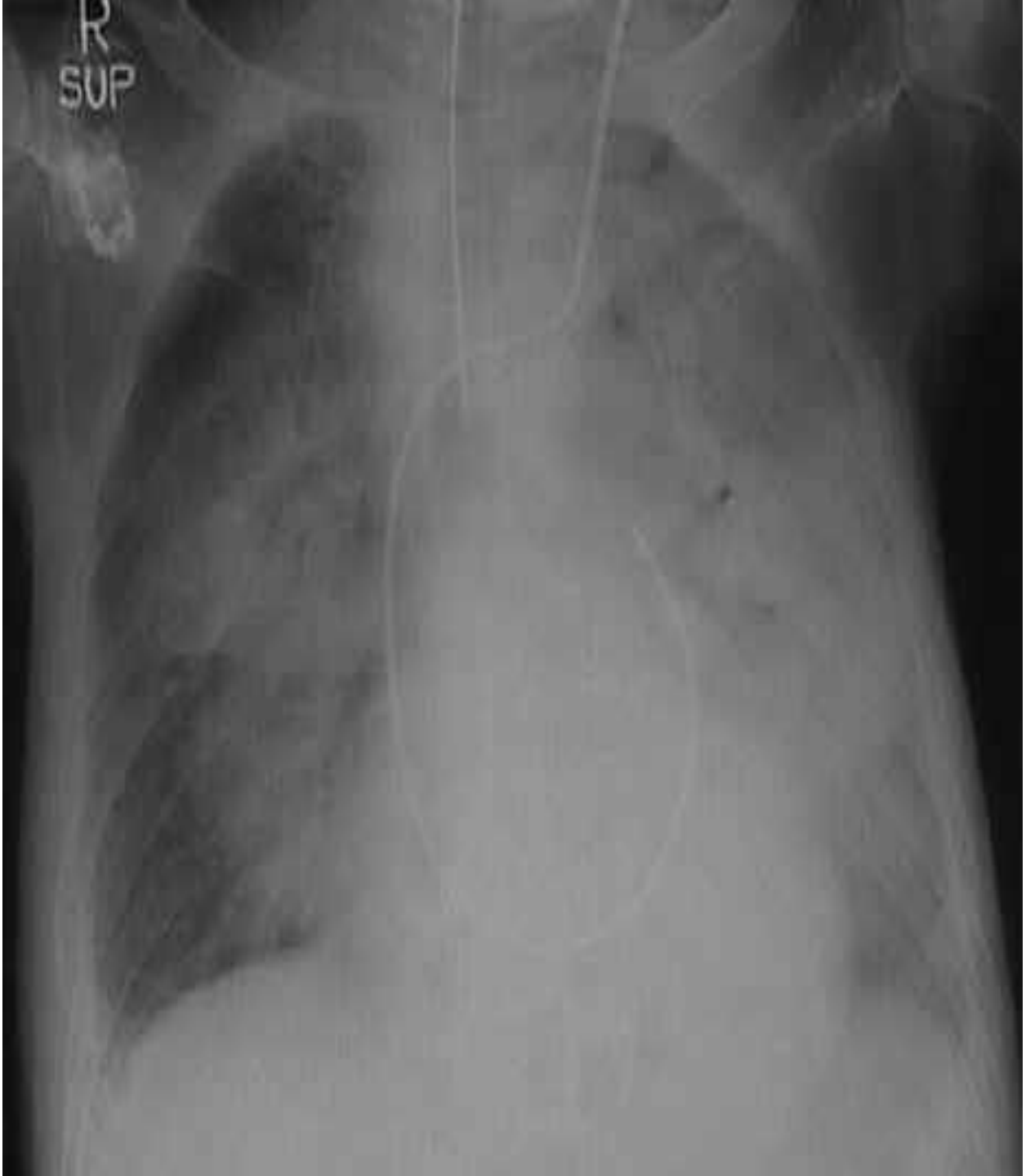
Activation of inflammatory mediators and cellular components resulting in damage to capillary endothelial and alveolar epithelial cells

Increased permeability of alveolar capillary membrane

Influx of protein rich edema fluid and inflammatory cells into air spaces

Dysfunction of surfactant


Normal Alveolus
Injured Alveolus during the Acute Phase
Alveolar air space
Type I cell
Epithelial basement membrane
Interstitium
Type II cell
Alveolar macrophage
Surfactant layer
Endothelial cell
Endothelial basement membrane
Capillary
Red cell
Fibroblast
Protein-rich edema fluid
Sloughing of bronchial epithelium
Necrotic or apoptotic type I cell
Inactivated surfactant
Activated neutrophil
Red cell
Leukotrienes
Oxidants
PAF
Proteases
Intact type II cell
Denuded basement membrane
Cellular debris
TNF-α, IL-1
Hyaline membrane
Alveolar macrophage
Fibrin
Migrating neutrophil
Proteases
IL-6, IL-10
MIF
TNF-α, IL-8
Widened, edematous interstitium
Procollagen
Gap formation
IL-8
IL-8
Platelets
Neutrophil
Swollen, injured endothelial cells
Fibroblast
Neutrophil


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# Clinical Features

Any patient having severe medical or surgical problem & the patient developed severe RF within the first 1 or 2 days, you should think about ARDS.

Again you find the features of the cause of the ARDS.

# Complications

- Bacterial super infection, because oedema fluid is a good medium for infection by many M.O.
- Multiple organ failure, because there will be septicemia which can leads to liver, kidney, heart, or brain failure that is why the mortality rate is high even in the best centers. Mortality rate in the best centers is about 50%, here in our country the mortality rate is about 90% !! because of lack of sophisticated equipment, the average mortality rate is about 70%.
- If the patient survive the acute phase then the patient will end up into pulmonary Fibrosis.

# Summery of finding

1) Progressive SOB (1-2 days) following any medical or surgical problem.

2) Central cyanosis (resistant to treatment).

3) Hypoxia & hypocapnia.

4) Wide-spread shadowing (bilateral) on chest X-ray.

5) On right sided cardiac catheterization to check the pulmonary venous pressure, it will be normal (i.e. there is normal pulmonary wedge pressure, yet there is pulmonary oedema.

6) Pulmonary fibrosis  is the sequel of those who survive .

7) Fine crepitation, due to pulmonary edema O/E

# Treatment

The patient should be cared in an "Intensive Care Unit" ICU, with facility for mechanical ventilator & Rx is directed toward:

A: cause (medical or surgical)

B: General measures

# General Measures

A- O2 therapy: Liberal therapy should be offered for all patient regardless of the cause; but not pure $O_2$ (high conc.) for more than 24h because it is irritant to the lung tissue.

B- Mechanical ventilation: in ARDS, there is damage to type -II pneumatocyte → decrease amount of surfactant, so you need always a positive pressure to keep the lung inflated & push the fluids away from the airspaces

The Mechanical ventilation are 2 types of assisted ventilation to keep this positive pressure :

1) *IPPV* “intermittent positive pressure ventilation"

2) *PEEP* "positive end expiratory pressure

# General Measures

**C- Other general measures that include**

1. Dealing with infection.
2. Feeding & IV nutrition may be needed in ICU
3. Cleanliness
4. Prevent bed sore
5. Electrolyte balance must be carefully controlled.
6. Use of Steroids

THANKS